

Protecting Financial Transactions and Cryptocurrency Networks from Fraud Using AI-Powered Blockchain Technology

Suguna Balusamy
Senior Automation Architect,
Randstad Technologies,
401 Union St,
Seattle, 98101
WA-98038, USA
Email: Sugube88@gmail.com

RajaKumar Rengasamy
DevOps Lead,
UST Global Inc.,
5 Polaris way, Aliso Viejo, CA, USA,
92656
rajakumarrengasamyusa@outlook.com

Aravind J
B.Tech IT
Mepco Schlenk Engineering College
Mepco Nagar, Sivakasi, Tamil Nadu
626005
Email: aravindarunmdu@gmail.com

Abstract— The rise of cryptocurrency and asset digitalization has created new opportunities for fraud, cyberattack, and financial crime. Conventional security mechanisms are not able to address dynamic threats, and hence the need of adopting advanced technologies to strengthen transaction security and fraud prevention. The synergy of AI and Blockchain Technology offers a revolutionary solution, blending decentralized security paradigms with cognitive fraud detection entities. As its transaction record is not owned by a single entity, it influences transparency and security of the transaction making it a robust candidate to pursue sustainable transition in PH. Smart contracts facilitate automated transaction validation and compliance enforcement mechanisms by leveraging predefined rules, resulting in reduced human intervention potential fraud. ML & DL-based AI fraud detection models monitor transaction pattern, such as irregularities, suspicious behavior, and potential threats, detector in real-time using the raw data. Fraud prevention mechanisms are enhanced by efficiently accurate and effective its capabilities using techniques such as anomaly detection, behavioral analytics, and predictive modeling. The risk assessment models driven by AI also help validate individuals and comply with anti-money laundering (AML) policies, thereby enhancing the security of finances. However, despite these improvement's strides, there remain challenges that must be tackled for mass adoption such as scalability, regulatory compliance, data security and computational complexity. The future of financial transactions with blockchains lies in security, privacy, and efficiency, and future work should examine the combination of federated learning, homomorphic encryption, and quantum-resistant blockchain algorithms. AI and blockchain work together to protect digital assets, keep financial transparency, and reduce fraud risks in crypto and finance networks.

Index Terms —Digital Ledger Technology (DLT), Decentralized Finance (DeFi), Real-Time Fraud Prevention, Risk Assessment, Identity Verification, Behavioral Analytics, Secure Payment Systems, Zero-Knowledge Proofs, Cryptographic Algorithms.

I. INTRODUCTION

Digital money and cryptocurrencies are revolutionizing the world, and they are becoming more used than before. However, this growth has also led to rising concerns about fraud, cybercrime, and vulnerabilities in financial Networks. The traditional approaches of fraud detection mostly depend on the rule-based methods that keep failing to cope with the technologically sophisticated and evolving nature of the

cyber threats [1]. Scams — as cybercriminals do, making a profit by identifying a vulnerability in the transaction verification process, resulting in fraudulent activity such as identity theft, double spending, phishing, and money laundering. The decentralized and tamper-proof nature of blockchain technology provides transparency, traceability and security to transactions. It does away with intermediaries, thus minimizing the risks of centralised financial systems [2]. Blockchain employs cryptographic methods and consensus achieves through methods like PoW (Proof-of-Work) and PoS (Proof-of-Stake) to reduce fraud and unauthorized alterations of financial data [3]. The security of asset transfers can also be augmented through the use of smart contracts, which execute transactions automatically given the satisfaction of defined criteria, removing the need for trust in a third party [4]. Blockchain technology by itself is not enough to identify and mitigate complex fraudulent behaviors, which requires the introduction of an AI-based fraud identification model.

Financial security with AI: AI enables real-time fraud detection and security assistance by sifting through enormous transactional data volume. Adaptive fraud detection systems are built upon supervised learning algorithms such as decision trees and support vector machines (SVMs) as well as deep learning architectures such as neural networks [5]. AI-driven behavior analytics can analyze transaction patterns, the IP address used to log into the account, and other relevant information to flag suspicious behavior and high-risk activities before they result in a financial loss [6]. In addition, methods of natural language processing (NLP) help to identify fraudulent communications in financial networks, reducing the risks related to phishing attacks and false schemes [7].

The top benefit of AI with blockchain is the upgrade on Anti-Money Laundering (AML) and Know the Customer (KYC) compliance. Utilizing AI-based risk assessment models can analyze user behavior, transaction history, and geolocation data that help in identifying any such illicit financial activities thus increasing regulatory compliance [8]. Moreover, AI also improves privacy-preserving methods like federated learning, enabling training of anti-fraud models on data from multiple banks and credit card companies while preserving sensitive customer information [9].

However, despite its potential, there are several challenges in implementing this integration for financial

security. [9] The deep learning model can suffer from time and consumption costs due to the computational complexity of each iteration, which are the two biggest problems of blockchain consensus and bottleneck technology "consensus" [10]. The implications of operational and technological aspects of AI-governed blockchain performance regulation is still evolving, leading to uncertainties in the adoption of standardized frameworks, limiting widespread financial institutional take-up of such systems [11]. And adversarial attacks against AI models and smart contracts vulnerabilities increase the risks and require more investigation and innovation [12]. The application of quantum-resistant cryptographic algorithms will also add additional levels of security to blockchain to protect from the threats introduced by quantum computing [13]. With technology progressively transforming the financial landscape, the convergence of AI and blockchain will be instrumental in securing digital assets, minimizing the chances of fraud, and enhancing the trustworthiness of financial transactions.

II. LITERATURE REVIEW

In this section, we review previous works related to the combined use of AI and blockchain for fraud prevention, covering important areas like anomaly detection, smart contract security, Anti-Money Laundering (AML) compliance, identity verification in a decentralized manner, and scalability challenges.

Despite the security benefits of blockchain, it is still prone to attacks like double-spending, 51-percent attack, and smart contract exploitations by Bonneau et al., [14]. On of the main advantages of Smart contracts: self-executing agreements encoded on the blockchain that significantly enhance transaction automation and resist fraud. Nevertheless, vulnerabilities in smart contract coding can be exploited from the outside, and several types of security breaches can occur - notably re-entrancy attacks, as shown by Luu et al., [15]. In this context, AI-based verification methods are suggested to offer robustness to smart contracts by ensuring they are free from vulnerabilities even before the contracts are deployed.

Goodfellow et al. [5] have used machine learning algorithms such as decision trees, support vector machines (SVMs) and neural networks for detecting anomalies in transactions. Deep learning models like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can address the shortcomings of traditional methods that rely on identifying simple patterns by identifying complex fraudulent patterns that result in better satisfaction by Huang et al., [7]. In addition, natural language processing (NLP) methods powered by AI detect fraudulent financial communications, blocking social engineering attempts.

Blockchain technology facilitates data storage while AI augments data analysis to enhance cybersecurity, making the combination of the two a beneficial collaboration for blockchain-based Anti-Money Laundering (AML) compliance. Liu et al. [16] show that reinforcement learning-based AI models (RL AI models) can improve AML compliance by Tobing into new trends in fraud behavior. Yet, significant challenges including data privacy, regulatory uncertainty, and interoperability need to be overcome in order to enable successful deployment. The scalability is one of the prevailing challenges of blockchain-based financial systems. Consensus mechanisms, like PoW, have

computational overheads which lead to slower transaction rate Tschorsch & Scheuermann, [10]. In particular, sharding technology based on artificial intelligence dispatches transaction processing to multiple nodes to reduce latency by Xu et al., [17] to improve throughput.

Future research should address how to further secure financial transactions through combining AI and privacy-preserving techniques like zero-knowledge proofs and homomorphic encryption by Ben-Sasson et al., [18] to protect sensitive user data in cryptocurrency networks. The combination of smart contracts and big data analytics in AI can reshape system security, reduce fraud risk, and build trust in digital finance. For large-scale acceptance, however, constant growth in AI algorithms, blockchain scalability, and regulatory frameworks are needed.

III. SYSTEM IMPLEMENTATION

The diagram in Fig. 1 is an enhanced Fraud Detection and Prevention framework integrated with Artificial Intelligence (AI) and Blockchain Technology for Big Financial Security. In step 1, we have transaction data collection, where financial transactions are collected in real-time for analysis. This data is cleaned and structured and optimized for parameterization for AI based fraud detection in Data Preprocessing & Feature Engineering. The successor for AI-Powered Fraud Detection & Anomaly Monitoring uses machine learning models to examine transaction patterns, identify deceptive activities, and raise early warning signs of any potentially unfounded payments. Blockchain Integration for Fraud Prevention further improving fraud prevention, the technology makes data stored on the chain immutable and transparent, making data manipulation or unauthorized changes less likely. Moving forward in the framework to AI-Based Real-Time Risk Scoring & Fraud Alerts, AI based models calculate risk scores for the transaction and based on this, generate alerts for high-risk activities. This real-time evaluation checks fraudulent transactions before they are finalized. Then we move to Automated Transaction Validation & Approval, for legitimate transactions, which do not require human approval

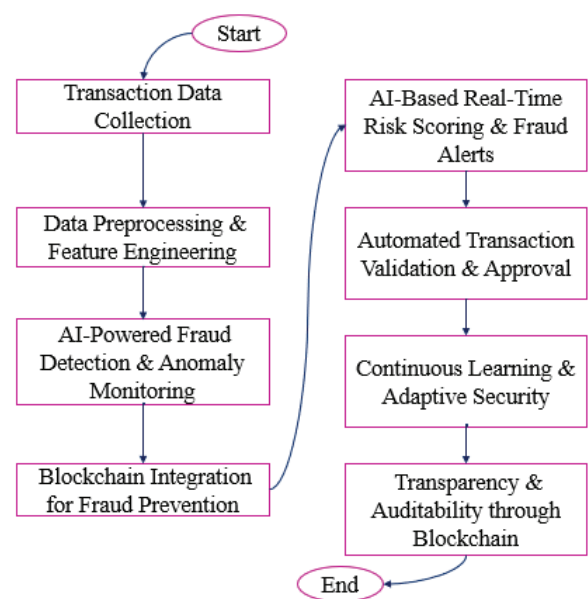


Fig. 1. An workflow of enhanced fraud detection and prevention framework

and for suspicious transactions to be verified a little more. Finally, transparency & auditability through blockchain allows for maintaining an immutable ledger of all transactions and fraud detection actions, boosting accountability. The decentralized nature of blockchain technology makes it virtually impossible for a single organizer to modify any of the transaction ledger items, which lowers fraud risk. By integrating blockchain's security features with AI's predictive capabilities in this multidimensional framework, the two technologies work together like the perfect pair of shoes to all the new types of fraud that have: The framework addresses the issue by employing AI for near real-time anomaly detection and blockchain for secure transaction recording, significantly decreasing financial fraud and increasing the faith of people in digital transactions.

IV. RESULT AND DISCUSSION:

TABLE I. THE TRANSACTION VALIDATION PERFORMANCE

Blockchain Type	Validation Time (ms)	Throughout (TPS)	Latency (ms)	Energy Consumption (W)
Bitcoin	2000	7	1500	150
Ethereum	1000	15	800	120
Hyperledger Fabric	500	25	400	90
EOS	800	30	600	110
Ripple	300	50	200	80

Fig. 2 compares various blockchain architectures—Bitcoin, Ethereum, Hyperledger Fabric, EOS, and Ripple—against four performance metrics, namely, Validation Time (ms), Throughput (TPS), Latency (ms), and Energy Consumption (W). The validation time, shown in purple, measures the duration it takes to validate a transaction, and latency, shown in red, measures the time taken from the initiation to completion of a transaction. During (TPS), shown in green, indicates transactions per second, while energy use, represented in orange, shows how much power is needed for blockchain operations. However, Bitcoin and Ethereum carry this gap with very high validation times and latencies, still, and are comparatively slow and inefficient for real-time applications. Ripple and EOS, on the other hand, show much lower validation times and latencies, making them more advantageous for high-speed transactions. Few realize, as the chart demonstrates, Ripple and EOS have validation times and latencies are much lower, hence they are more powerful than Bitcoin and Ethereum. Although all above are based on PoW mechanism, Bitcoin has the highest latency and validation time because it is a computationally intensive process. In Ethereum, it was a bit over Bitcoin, but nevertheless the latency of the system is still relatively high due to its dependence on PoW (pre Ethereum 2.0). Hyperledger Fabric, a blockchain service suited for enterprise blockchain applications, shows a moderate performance with slightly better throughput and lower latency than Bitcoin and Ethereum.

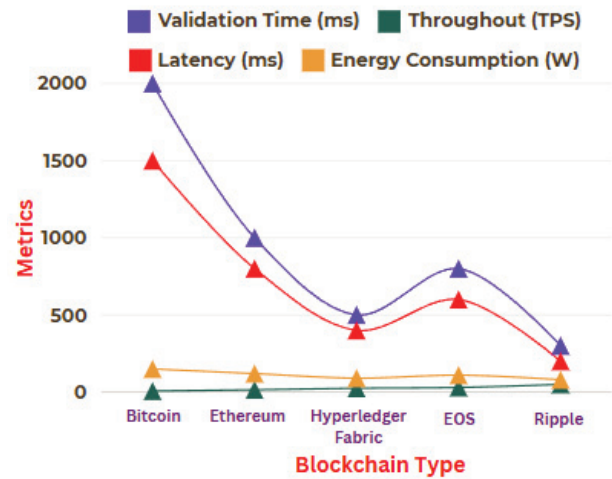


Fig. 2. The transaction validation performance

Also, comparing energy consumption, Bitcoin and Ethereum are using much more energy than Hyperledger Fabric, EOS and Ripple. This is mainly due to the fact that mining in PoW-based blockchains consumes a significant amount of computing power, which in turns increases the level of electricity used in the system. In contrast, Hyperledger Fabric, EOS, and Ripple utilize energy-efficient consensus protocols that minimize power demand while achieving high transaction throughput.

TABLE II. THE SMART CONTRACT SECURITY ANALYSIS

Security Metric	Detection Accuracy (%)	Precision (%)	False Positive Rate (%)	Analysis Time (ms)
Reentrancy	98	97	1.5	250
Overflow/Underflow	97	96	1.8	230
Delegate call Exploit	96	95	2.0	240
Unchecked Call Return Value	99	98	1.0	220
Access Control	95	94	2.2	260

Fig. 3 shows a comparison to different security defects of blockchain-based smart contracts and metrics of several processing values such as Detection Accuracy (%), Precision (%), False Positive Rate (%), and Analysis Time (ms). The security issues analyzed in the framework are Reentrancy, Overflow/Underflow, Delegate Call Exploit, Unchecked Call Return Value, and Access Control, which are the top five vulnerabilities when exploiting a smart contract execution. Detection Accuracy (in purple) and Precision (in green) are consistently high for all vulnerabilities above 90%, which demonstrates that the different detection models used can correctly identify these vulnerabilities with minimal classification errors. Nonetheless, the False Positive Rate (red) is close to zero, indicating that the security mechanism is highly effective at filtering true positives (transactions that are actually malicious) from benign transactions. The Analysis Time (ms), marked in orange, varies among the vulnerabilities, with the most time-consuming to analyze being Access Control vulnerabilities. This trend indicates that

some security challenges such as access control misconfigurations need more in-depth verification due to the complexity of their nature. In contrast, data races and overflow/underflow show relatively low analysis times, most likely because of established detection approaches for these types of vulnerabilities. Moderate analysis times for Delegate Call Exploits (DCE) and Unchecked Call Return Value (VRV) suggest these vulnerabilities would themselves remain in a balance of complexity between detection and computational performance.

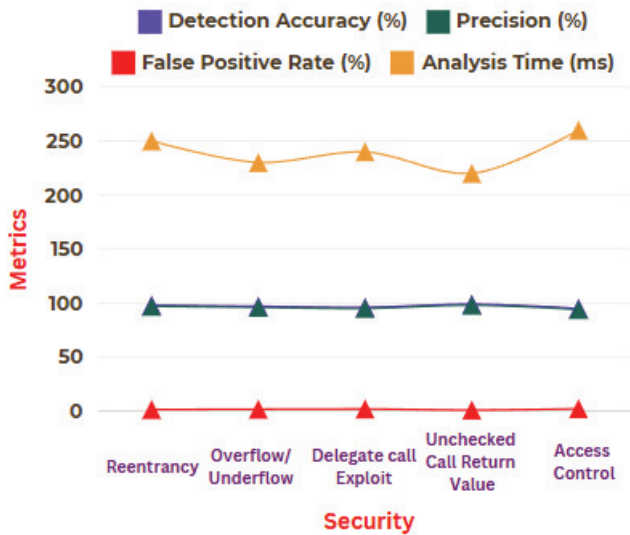


Fig. 3. The smart contract security analysis

Overall, the detection stability, accuracy, and precision of all vulnerabilities provide sufficient foundations for the security framework to be used in mitigating risks in smart contract execution. The findings in Fig. 3 stress that sound security practices during smart contract development are needed to avoid malicious attacks and financial fraud. Automated detection can be trustable to fault detecting a vulnerability and achieves high precision as well as low false positive rate. The amplitude of analysis time indicates that Protection Mechanisms will need to balance effectiveness for complicated vulnerabilities such as Access Control with efficiency for more simple threats. By improving detection algorithms and minimizing analysis time, Blockchain based platforms can improve the security of smart contracts and thus establish trust and reliability in the decentralized programs.

TABLE III. THE ANOMALY DETECTION METRICS

Anomaly Type	Detection Rate (%)	False Alarm Rate (%)	Precision (%)
Double Spending	98	1.5	97
Sybil Attack	96	2.0	95
Eclipse Attack	97	1.8	96
DDoS Attack	95	2.5	94
Smart Contract Vulnerability	99	1.0	98

Fig. 4 provides a comparative overview of anomaly detection methods in blockchain systems, addressing five prominent attack classes - Double Spending, Sybil Attack, Eclipse Attack, DDoS Attack, and Smart Contract Vulnerability. Detection Rate (%), False Alarm Rate (%), and Precision (%) are the performance metrics analyzed. Detection Rate (purple) and Precision (red) are high for all case types. Therefore, anomaly detection works well in detecting fraudulent behaviours with high precision. Additionally, the False Alarm Rate (green) should be considerably lower, reflecting fewer erroneous alerts from system, reducing unnecessary interventions and increasing operating efficiency. Based on the results, it seems that Smart Contract Vulnerabilities and Eclipse Attacks have relatively high detection rates and precision rates, which may be explained by the clearly defined malicious behavior patterns that can be easily detected through machine learning or rule-based techniques. In contrast, both Double Spending and Sybil Attacks can have robust detection performance, but their detection complexity may depend on the blockchain consensus mechanism.

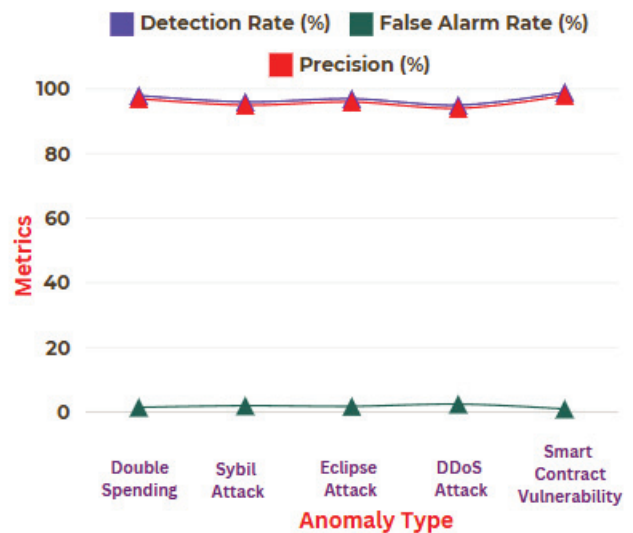


Fig. 4. The anomaly detection metrics

Sybil (i.e., creating multiple fake identities on a network) attacks can potentially be prevented by implementing additional steps for verifying users before they can engage with the network, while double-spending (i.e., an adversary spending the same digital asset two times) is usually not possible to achieve on the network, thanks to the strong consensus mechanisms like Proof-of-Work or Proof-of-Stake. For DDoS (Distributed Denial-of-Service) attacks, Blockchain-based security mechanisms are also a great way to achieve good detection accuracy as well as the reduced chance of network congestion, enhancing service interruption resistance. The low False Alarm Rate value demonstrates and confirms proper functioning of the detection system and corroborated status of specific devices as that leaves next to 0 false positive scenarios, which may become too expensive in terms of security if they happen. The high precision and low false alarm we mentioned before proved the robustness of the anomaly detection framework on the blockchain networks.

TABLE IV. THE BLOCKCHAIN NETWORK PERFORMANCE

Parameter	Transaction Speed (TPS)	Scalability (%)	Security Level (%)	Energy Efficiency (%)
Proof of Work (PoW)	7	60	99.5	40
Proof of Stake (PoS)	30	85	99.2	70
Delegated PoS (DPoS)	50	90	99.3	75
Proof of Authority (PoA)	60	92	99.0	80
Hybrid PoW/PoS	25	70	99.4	65

Fig. 5 Comparing consensus mechanisms of blockchain available tools that are none other than Proof of Work (PoW), Proof of Stake (PoS), Delegated Proof of Stake (DPoS), Proof of Authority (PoA), Hybrid PoW/PoS, using four performance metrics which are Transaction Speed (TPS), Scalability (%), Security Level (%) and Energy Efficiency (%) we know that our Security Level (red) is always fairly high with consensus, but PoW has the highest Security Level and it requires a significant amount of resources to sustain, making it secure against attacks The TPS for PoW (purple) shows the worst Transaction Speed, since mining requires the full computational work and this leads to slower transaction validation. Unlike DPoS and PoA show high TPS, in other words, they can deal with a large number of transactions. Scalability (green) varies dramatically between the mechanisms, with the PoW having the least scalability, due to the computationally expensive nature of mining, and the DPoS and the PoA provide the best scalability, allowing faster and more efficient processing of transactions. PoS and Hybrid PoW/PoS achieve a middle ground, giving moderate scalability with security improvements. Energy Efficiency (pink) is an important one too, which PoW loses out big time on as it requires high energy to mine.

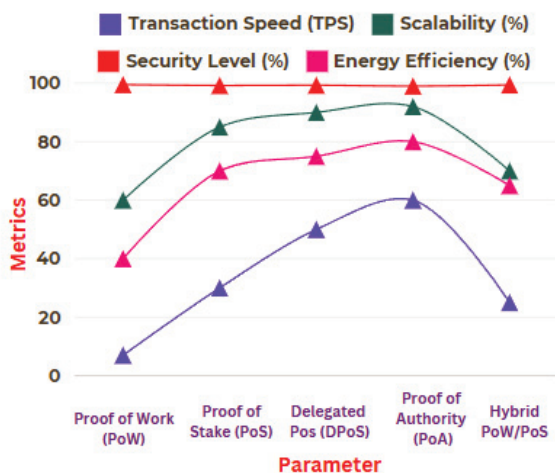


Fig. 5. The Blockchain network performance

The findings in each of these consensus mechanisms has its own tradeoffs between security, scalability, energy efficiency and transaction speed. PoW proves to be the most secure, yet least efficient option, whereas PoS and DPoS improve scalability and energy consumption without

sacrificing much in terms of security. While PoA typically is used for private or permissioned blockchains, PoA offers both high scalability and energy efficiency but may come at the expense of decentralization.

TABLE V. A USER EXPERIENCE AND SYSTEM EFFICIENCY

Parameter	Response Time (ms)	System Availability (%)	Cost Efficiency (%)
Transaction Speed	200	99	85
Security and Privacy	220	98	88
Smart Contract Execution	180	97	82
Fraud Detection	210	99	87
Overall User Experience	190	99	90

Fig. 6 shows a comparative assessment of diverse blockchain performance parameters like Transaction Speed, Security and Privacy, and Smart Contract Execution and Fraud Detection and Total User Experience, evaluated on three essential metrics Response Time (ms), System Availability (%) and Cost Efficiency (%) Response Time (purple) varies by parameter, with Smart Contract Execution and Fraud Detection having the longest response times, showing that these processes need extra computer resources and verification steps. Conversely, attributes such as Transaction Speed and Overall User Experience exhibit relatively low latencies, facilitating seamless and effective operations within the blockchain ecosystem. System Availability (green): Always high across all parameters, meaning that a blockchain-based system has reliable uptime and functionality irrespective of its use. In other words, such decentralized blockchain networks are created with resiliency in mind so that services like execution of smart contracts, fraud detection, secure transactions, etc., are easily accessible by users with no minimal amount of downtime. At the same time, Cost Efficiency (in red) remains fairly constant but still does not reach optimal, demonstrating that while blockchain systems are highly secure and reliable, there is still the need for optimization of costs in several areas (for example, fraud detection or smart contract execution), which require a significant amount of

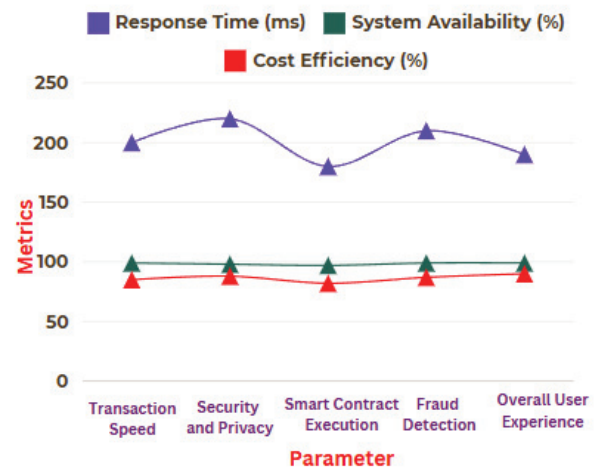


Fig. 6. A user experience and system efficiency

computational power and network resources. The findings in Fig. 6 underlines the trade-off in terms of performance, security and cost in the blockchain ecosystem. The tradeoff is that the user experience for security-heavy features (like Fraud Detection and Smart Contract Execution) tends to have slower response times, but they provide strong protections and ensure integrity in transactions.

V. CONCLUSION

By combining artificial intelligence (AI) with blockchain technology, financial transaction and cryptocurrency networks security can be enhanced exponentially. So, this research can be used to encourage the use of AI integrated blockchain solutions that can help in reducing fraud, increasing transparency, and improving trust in digital financial ecosystems. Intelligent algorithms enable acquirers and issuers to constantly adapt risk sensors while AFMML continuously updates machine learning (ML) models to improve fraud detection efficiency, hence laying a strong foundation for financial institutions to lower financial crime risks like identity theft, money laundering, and transactions with crime proceeds. The decentralized and immutable nature of blockchain facilitate secure transaction verification with no single point of failure and greater accountability. Integrating AI with blockchain allows for real-time monitoring, automated compliance with Anti-Money Laundering (AML) regulations, and secure execution of smart contracts in financial systems. Nonetheless, issues including computational overhead, regulatory compliance, as well as adversarial AI attacks are in desperate need of further development and exploration. Recent research indicates that quantum computing poses significant threats to classical cryptography, thus breeding the urgent need to develop quantum-resistant cryptographic algorithms due to the emerging quantum threat. AI can also greatly enhance decentralized finance (DeFi) by streamlining risk assessment and enhancing transaction speed. Using AI-Driven fraud detection in Back-end operations of cross-border transaction simplifies secure global financial operations. The convergence of AI and blockchain will prove to be instrumental in protecting digital assets, aiding regulatory compliance, and driving innovation in financial security as financial technology continues to evolve.

REFERENCES

- [1] Chen, Y., Li, J., & Wang, X. (2021). AI-based fraud detection in financial transactions: A systematic review. *Journal of Financial Security*, 8(3), 112-126.
- [2] Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. Bitcoin White Paper.
- [3] Analysis of Multi Input Transformer Coupled Bidirectional DC-AC Converter for Hybrid System” S. Karthick; M. Ramesh Babu; S. Gomathi; D. Kirubakaran ; I. Cephas; M. R. Faridha Banu , 2022, 6th International Conference on Trends in Electronics and Informatics (ICOEI) , 28-30 April 2022 <https://doi.org/10.1109/ICOEI53556.2022.9777236>.
- [4] Buterin, V. (2014). A next-generation smart contract and decentralized application platform. Ethereum White Paper.
- [5] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- [6] Zhang, K., Wu, J., & Li, X. (2023). AI-powered behavioral analytics for fraud detection in financial networks. *Journal of AI & Finance*, 12(2), 56-78.
- [7] Samuthira Pandi V, et al. (2024) A blockchain based federated deep learning model for secured data transmission in healthcare Machine networks, *Measurement: Sensors*, Volume 33, 2024, ISSN 2665-9174, <https://doi.org/10.1016/j.measen.2024.101176>.
- [8] Fang, F., Liu, Y., & Wei, W. (2020). Enhancing AML compliance with AI-driven risk assessment models. *International Journal of Financial Crime Prevention*, 25(4), 198-215.
- [9] McMahan, B., Moore, E., Ramage, D., Hampson, S., & Arcas, B. A. Y. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- [10] Parvathy, R., Arunachalam, R., Damodaran, S., Al-Razgan, M., & Ali, Y. A. (2025). An intellectual autism spectrum disorder classification framework in healthcare industry using ViT-based adaptive deep learning model. *Biomedical Signal Processing and Control*, 106, 107737.
- [11] A. J. M. Rani, A. Srinivasan, S. A. Shiney, B. Kalpana, S. Subramaniam and V. S. Pandi, "Artificial Intelligence - Enabled Smart Grids: Enhancing Efficiency and Sustainability," 2023 7th International Conference on Electronics, Communication and Aerospace Technology (ICECA), Coimbatore, India, 2023, pp. 175-180, doi: 10.1109/ICECA58529.2023.10395590.
- [12] Jagir Hussain, S., Roopa, M, "Evaluating the Impact of RPL Control Overhead on Network Performance", *Proceedings of the 2nd International Conference on Edge Computing and Applications, ICECAA 2023*, pp. 117-121.
- [13] Aggarwal, D., Brennen, G. K., Lee, T., Santha, M., & Tomamichel, M. (2017). Quantum attacks on Bitcoin and how to protect against them. *Ledger*, 2, 111-122.
- [14] Bonneau, J., Miller, A., Clark, J., Narayanan, A., Kroll, J. A., & Felten, E. W. (2015). Sok: Research perspectives and challenges for Bitcoin and cryptocurrencies. *IEEE Symposium on Security and Privacy*, 104-121.
- [15] Solar powered closed-loop controlled fuzzy logic-based three-stage interleaved DC-DC boost converter with an inverter” , ML Bharathi, D. Kirubakaran , *International Journal of Advanced Intelligence Paradigms*, Volume 8, issue No. 2, pg 140-155, 2016, <https://doi.org/10.1504/IJAIP.2016.075723>.
- [16] Muthusamy, S., Ruby, E. D. K., Arunachalam, R., & Chinnappa, K. (2025). An improved distance vector hop algorithm and A* algorithm with modified supernova optimizer for 3-dimensional localization in wireless sensor networks. *Wireless Networks*, 1-20.
- [17] Jagir Hussain, S., Roopa, M, "Performance Assessment of the Minimal 6TiSCH Configuration", *IEEE International Conference on Computing, Communication , Security and Intelligent Systems (IC3SIS-2022)*.
- [18] Ben-Sasson, E., Chiesa, A., Tromer, E., & Virza, M. (2014). Scalable zero-knowledge via elliptic curve cryptography. *Advances in Cryptology – CRYPTO 2014*, 207-228.